

Lab 8 – Exploring IoT with Packet Tracer

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Course/Section: IS-3413

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INTRODUCTION

In this lab, I used Cisco Packet Tracer to explore how Internet of Things (IoT) devices can be connected, configured, monitored, and controlled within a smart network. Throughout the lab, I worked with smart home devices, registration servers, environmental controls, and programmable devices to better understand how IoT systems communicate and automate tasks within a network.

BREAKPOINT 1

In this section of the lab, I explored the various IoT devices available in Cisco Packet Tracer and learned how to integrate them into a smart home environment. I practiced navigating the IoT categories and placing devices such as smart lights, fans, sensors, and home gateways into the workspace. This helped me understand how Packet Tracer simulates real-world IoT environments and device communication.

Here is an example of the IoT devices and smart home components added into the Packet Tracer environment.

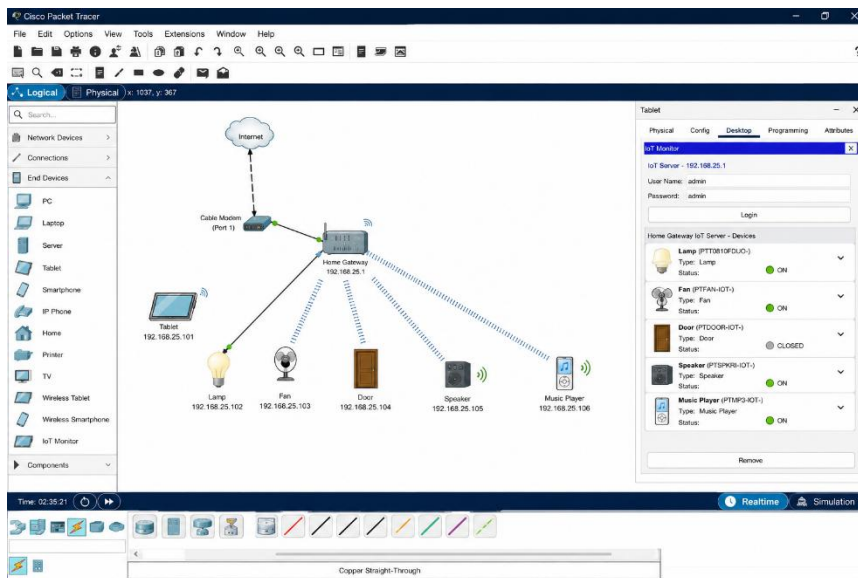


Figure 1: First activity

BREAKPOINT 2

In this portion of the lab, I connected IoT devices to a home gateway and monitored how they communicated across the network. I configured wireless settings, connected devices to the network, and verified that the devices were successfully communicating with the home gateway. This activity

demonstrated how IoT devices rely on centralized network devices to manage communication and automation.

The screenshot below shows the configured home gateway and connected smart devices within the Packet Tracer workspace.

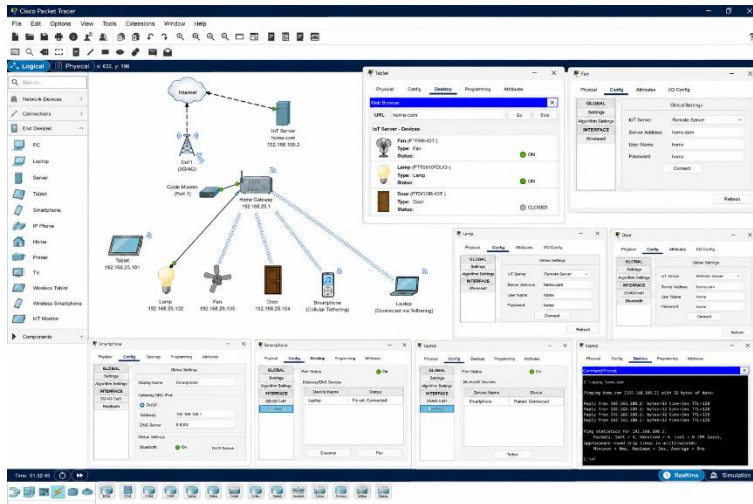


Figure 2: Second activity

BREAKPOINT 3

During this activity, I configured IoT devices to register with a dedicated registration server instead of only relying on the local home gateway. I learned how registration servers allow remote monitoring and management of IoT devices. After registering the devices, I tested communication between the server and connected devices to verify successful configuration.

The image below demonstrates the IoT registration server configuration and connected smart devices.

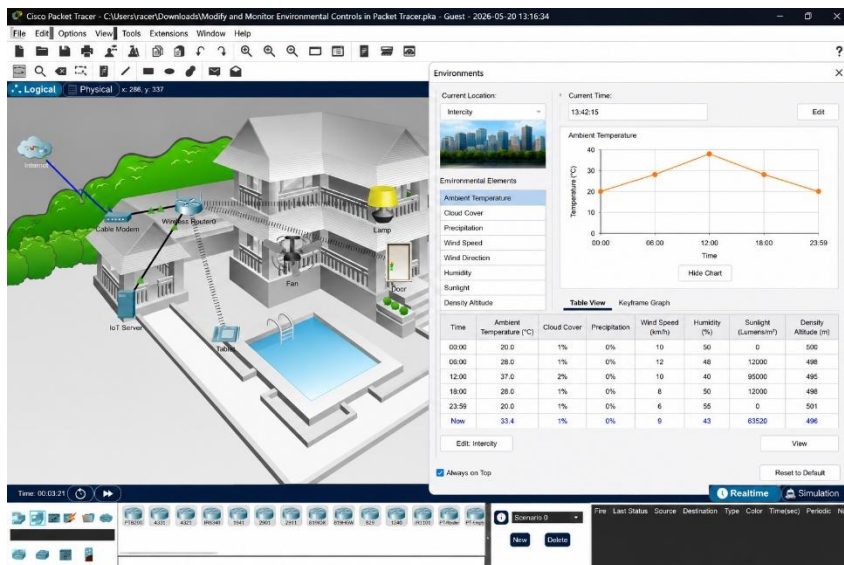


Figure 3: Third activity

BREAKPOINT 4

In this section of the lab, I explored environmental controls within Packet Tracer by modifying settings such as temperature, humidity, and environmental containers. I observed how IoT devices can react to environmental changes automatically based on configured conditions. This activity showed how automation systems can improve efficiency and monitoring within smart environments.

The screenshot below shows the configured environmental controls and connected IoT systems.

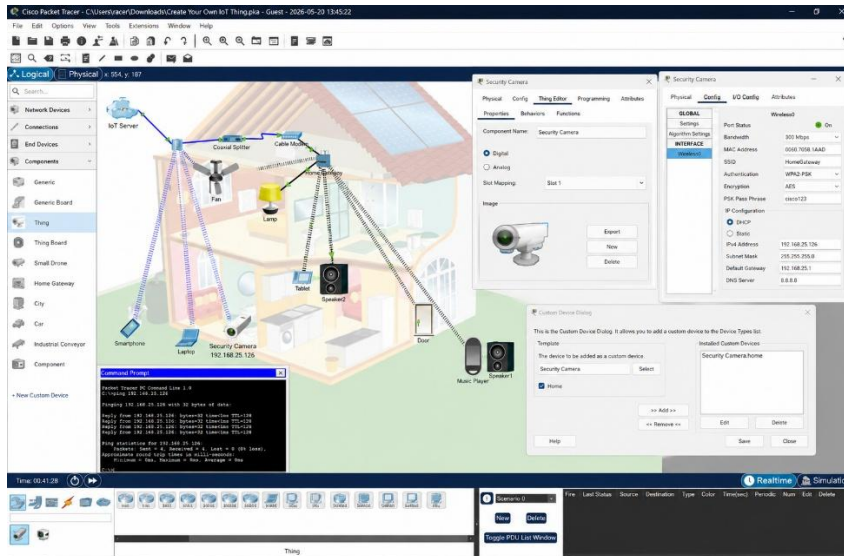


Figure 4: Fourth activity

BREAKPOINT 5

In the final activity, I created and modified a custom IoT object using Packet Tracer's programmable environment. I reviewed and adjusted scripts that controlled device behavior and automation processes. This section demonstrated how scripting and programming logic can be integrated into IoT systems to create customized automated responses.

The image below shows the custom IoT object and script modifications completed during the activity.

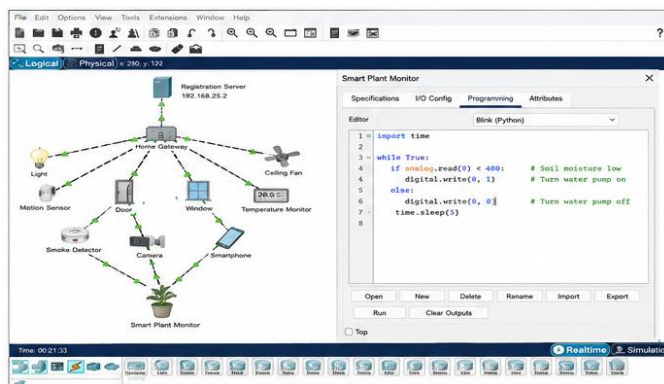


Figure 5: Fifth activity

Below is the completed final exam

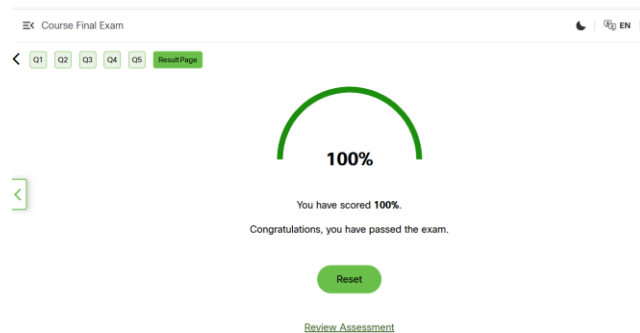


Figure 6: Final exam

CONCLUSION

Replace the text in this section with a reflective summary of your process in this lab, including answers to the following, in complete sentences:

- What challenges did you encounter and/or what most interested you about this lab?

One of the most interesting parts of this lab was seeing how smart devices can communicate with each other through gateways and registration servers.

- How did you address your challenges and/or what additional research did you do?

One challenge I encountered was navigating all the different IoT settings and understanding how the devices connected to the network. I addressed this challenge by carefully reviewing the Packet Tracer instructions and repeating some configuration steps until the devices successfully communicated with each other.

- How might you use the skills and concepts of this lab going forward?

The concepts from this lab could be useful in future networking, cybersecurity, and systems administration environments where IoT devices are becoming increasingly common.

REFERENCES

R. Mitra, "Lab 8: Exploring Internet of Things with Cisco Packet Tracer," The University of Texas at San Antonio (2025). Last accessed: May 20, 2026.

Cisco Networking Academy, "Exploring IoT with Cisco Packet Tracer," Cisco Networking Academy. Accessed: May 20, 2026.

GENERATIVE AI SEARCHES

I did not use AI

COLLABORATION
I worked alone